

SIMATS ENGINEERING

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0404

CO 3 – ASSESSMENT TOOL 2: DATA INTERPRETATION

COURSE OUTCOME COVERED

- Apply data preparation and exploratory data analysis techniques, including data summarization, data distribution analysis, outlier treatment, and measures of asymmetry, to preprocess and interpret datasets effectively.
- Analyze datasets using descriptive statistical measures such as mean, variance, sampling methods, covariance, and correlation to identify patterns, relationships, and support data-driven decision-making.

Question 11: Standardization/Normalization

A feature Age ranges from 18–75 and another feature Income ranges from ₹20,000–₹20,00,000 in a dataset meant for a distance-based algorithm. Explain, with reasoning, why failing to standardize these variables would distort analysis, and calculate the z-score for Age = 45 if mean = 40 and SD = 12.

Answer:

Distance-based algorithms such as K-Nearest Neighbors and K-Means compare observations using distances. If the variables are not standardized, the feature with the much larger numerical scale can dominate the distance calculation.

Here, Age varies over roughly 57 units, whereas Income varies over about ₹19,80,000. Therefore, a difference in income can contribute a vastly larger numerical distance than a meaningful difference in age. The algorithm may consequently treat income as much more important simply because of its scale, not because it is actually more informative.

Standardization puts variables on a comparable scale, allowing each feature to contribute according to its relative deviation rather than its original units.

Z-score formula: $z = (X - \mu) / \sigma$

For Age = 45, mean $\mu = 40$ and SD $\sigma = 12$:

$$z = (45 - 40) / 12 = 5 / 12 = 0.4167 \approx 0.42.$$

Final Answer: The Age value 45 has a z-score of approximately 0.42, meaning it is about 0.42 standard deviations above the mean. Without standardization, Income would disproportionately influence distance calculations.

Question 12: Distribution Shape from Summary Statistics Alone

A dataset has Mean = 45, Median = 45, Mode = 45, skewness ≈ 0 , but kurtosis = 5.8 (leptokurtic). Explain what this combination tells you about the distribution's shape that skewness and central tendency alone would miss.

Answer:

Mean = Median = Mode = 45 and skewness ≈ 0 indicate that the distribution is approximately symmetric and has no substantial left or right skew.

However, these measures do not describe how concentrated the observations are around the center or how heavy the tails are. The kurtosis value of 5.8 indicates a leptokurtic distribution, meaning it has heavier tails and a sharper concentration around the center compared with a normal distribution.

Thus, two distributions can have the same mean, median, mode, and approximately zero skewness while having very different tail behavior. In this case, extreme observations are more likely than they would be in a normal-shaped distribution.

Final Answer: The distribution is approximately symmetric and centered at 45, but it is more peaked and has heavier tails than a normal distribution. Kurtosis reveals the tail and peak behavior that central tendency and skewness alone cannot show.

Question 13: Grouped Data — Mean and Variance

Class intervals are 0–10, 10–20, 20–30, 30–40, and 40–50 with frequencies 5, 8, 15, 12, and 10. Calculate the mean and variance using the step-deviation method. Show the assumed midpoints and working.

Answer:

Step 1: Find the class midpoints. Choose assumed mean $A = 25$ and class width $h = 10$.

Class Interval	Frequency (f)	Midpoint (x)	$u = (x - A)/h$	fu	fu^2
0–10	5	5	-2	-10	20
10–20	8	15	-1	-8	8
20–30	15	25	0	0	0
30–40	12	35	1	12	12
40–50	10	45	2	20	40
Total	50			14	80

Step 2: Calculate the mean using the step-deviation formula.

$$\text{Mean} = A + h(\Sigma fu / \Sigma f) = 25 + 10(14 / 50) = 25 + 2.8 = 27.8.$$

Step 3: Calculate variance using the step-deviation formula.

$$\begin{aligned}\text{Variance} &= h^2[(\Sigma fu^2 / N) - (\Sigma fu / N)^2] \\ &= 10^2[(80 / 50) - (14 / 50)^2] \\ &= 100[1.6 - 0.0784] = 100(1.5216) = 152.16.\end{aligned}$$

Final Answer: Mean = 27.8; Variance = 152.16 square units.

Question 14: Comparing Two Distributions via Coefficient of Variation

Portfolio A has mean return = 12% and SD = 3%. Portfolio B has mean return = 18% and SD = 5%. An investor claims Portfolio B is riskier in absolute terms but not necessarily worse per unit of return. Evaluate this claim quantitatively using Coefficient of Variation and explain what CV captures that SD alone does not.

Answer:

Coefficient of Variation (CV) = (SD / Mean) × 100.

Portfolio	Mean Return	SD	CV
A	12%	3%	$(3/12) \times 100 = 25\%$
B	18%	5%	$(5/18) \times 100 \approx 27.78\%$

Portfolio B has the larger standard deviation (5% compared with 3%), so it is riskier in absolute terms.

However, CV measures risk relative to the average return. Portfolio A has CV = 25%, while Portfolio B has CV ≈ 27.78%. Therefore, B also has slightly higher risk per unit of return.

CV captures relative variability by scaling the standard deviation according to the mean. SD alone measures absolute dispersion and does not account for the size of the average return.

Final Answer: The investor is partly correct: Portfolio B is riskier in absolute terms, but quantitatively it is also slightly worse in risk per unit of return because its CV (27.78%) is higher than A's CV (25%).

Question 15: Outlier's Effect on Variance

A dataset of 10 values has variance = 25. One value, originally 40 (close to the mean of 42), is mistakenly recorded as 400. Without recalculating exactly, explain directionally and reason about the magnitude of distortion to (a) the mean and (b) the variance — and why variance is affected disproportionately more than the mean.

Answer:

(a) Effect on the mean: The recorded value increases from 40 to 400, an increase of 360. Since there are 10 observations, the arithmetic mean would increase by approximately $360/10 = 36$ units, assuming the other nine observations remain unchanged.

The original mean was 42, so the new mean would be approximately 78. The exact value is not required for the directional explanation.

(b) Effect on the variance: The variance would increase dramatically. The original value 40 was close to the mean of 42, so its contribution to squared deviations was small: $(40 - 42)^2 = 4$.

After the error, the value becomes 400. Its distance from the original mean is about 358, giving a squared deviation of approximately $358^2 = 128,164$. Even after accounting for the shift in the mean, this single extreme value contributes a very large amount to the total squared deviations.

Variance is disproportionately affected because deviations are squared. A large error therefore grows much faster in the variance calculation than in the mean calculation. The mean changes linearly with the erroneous value, whereas variance gives extreme deviations a quadratic effect.

Final Answer: (a) Mean increases substantially, by about 36 units. (b) Variance increases enormously compared with the original variance of 25. This happens because variance squares deviations, making it highly sensitive to extreme outliers.

SUMMARY OF FINAL ANSWERS

Question	Final Answer
11	Age z-score ≈ 0.42 ; standardization prevents Income from dominating distance calculations due to scale.
12	Approximately symmetric distribution centered at 45, but leptokurtic with heavier tails and greater peak concentration.
13	Mean = 27.8; Variance = 152.16 square units.
14	CV(A) = 25%; CV(B) $\approx 27.78\%$. B has greater absolute and relative variability.
15	Mean increases by about 36 units; variance increases dramatically because squared deviations magnify the outlier's effect.

